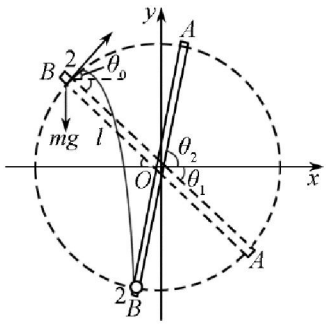
of the rod to the horizontal be θ_1. Then $f_1(\theta_1)=\mu N_1(\theta_1)$,



that is, $\frac{10}{3}mg\sin\theta_1=\frac{3}{4}\mu mg\cos\theta_1$,

(16)

Substituting the data into (16) gives

$\theta_1=\frac{\pi}{6}$。

(17)

At $\theta=\theta_1$, ball 1 starts sliding outward. Ball 2 starts next to the end, so ball 2 leaves after negligible time. Thus the rod's angle when ball 1 leaves is also $\theta_1=\frac{\pi}{6}$。

When ball 1 leaves, the massless rod exerts no force on ball 2. Ball 2 then feels gravity alone. By (2007), its initial speed in projectile motion is

$$v_0=l\sqrt{\frac{2g\sin\theta_1}{3l}}=\sqrt{\frac{3gl}{3}}。\qquad(18)$$

The initial velocity makes the following angle with the horizontal:

$$\theta_0=\frac{\pi}{2}-\theta_1=\frac{\pi}{3}。\qquad(19)$$

Ball 2 stays on the rod during flight; neither exerts a force. The rod follows without affecting it. Ball 2 leaves when its distance from the pivot is again l. Take the axes Oxy shown. At time t, taking the start of ball 2's flight as time zero, ball 2 has coordinates

$$x=-l\cos\theta_1+v_0\cos\theta_0 t,\qquad(20)$$

$$y=l\sin\theta_1+v_0\sin\theta_0 t-\frac{1}{2}gt^2,\qquad(21)$$

When ball 2 leaves the rod,

$$l^2=x^2+y^2。\qquad(22)$$

Using (17)(18)(19) gives

$$t^2\left(t^2-2\sqrt{\frac{l}{g}}t-\frac{2}{3}\frac{l}{g}\right)=0,\qquad(23)$$

Solving gives

$$t=\left(1+\frac{\sqrt{15}}{3}\right)\sqrt{\frac{l}{g}}。\qquad(24)$$

At this time,